Letharia vulpina (L.) Hue

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Letharia vulpina is a lichenised fungus part of the family of Parmeliaceae; it is a fruticose lichen with a coral-like shape usually pendulous, recognizable thanks to its bright yellow to green colour given by the vulpinic acid, substance that can be found in the cortex of the thallus, and is usually found on the bark or wood of dead or alive conifers. Its life cycle is characterized mainly by a vegetative reproduction through the formation of soredia, granules made of an algal cell covered by some hyphae of the fungus, and sometimes also isidia, small fragment of the thallus that detach with both the fungus and the alga covered by the cortex of the lichen (this enables the dispersal of the two organisms together ensuring germination, if the soredia or the isidia reach a suitable substrate); sexual reproduction have been very rarely observed through the production of apothecia. Like every lichen, it is characterized by a symbiosis of a mycobiont and a photobiont, in this case a green algae of the genus *Trebouxia*; it is still discussed whether the relationship between the two organisms is mutualistic, since the photobiont can live independently from the mycobiont, who on the other hand needs the symbiosis to survive in the environment. Another common feature with other lichens is its high tolerance to extremely low temperatures, being able to reactivate its metabolism rapidly and to being active (photosynthetic) even at – temperatures.

Taxonomy and classification:

Kingdom: Fungi
Phylum: Ascomycota
Class: Lecanoromycetes
Order: Lecanoroales
Family: Parmeliaceae
Genus: *Letharia*
Species: *L. vulpina*

Fun fact:
This lichen is also known as the “wolf lichen” (even if *vulpina* derives from the Latin word vulpine which means fox) since in the past it was used by humans to poison wolves and foxes; the vulpinic acid contained in the lichen – also used as a pigment by some communities – was added to meat or carcasses with glass shards so that when those animals started to eat the baits/preys, they would get poisoned.

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